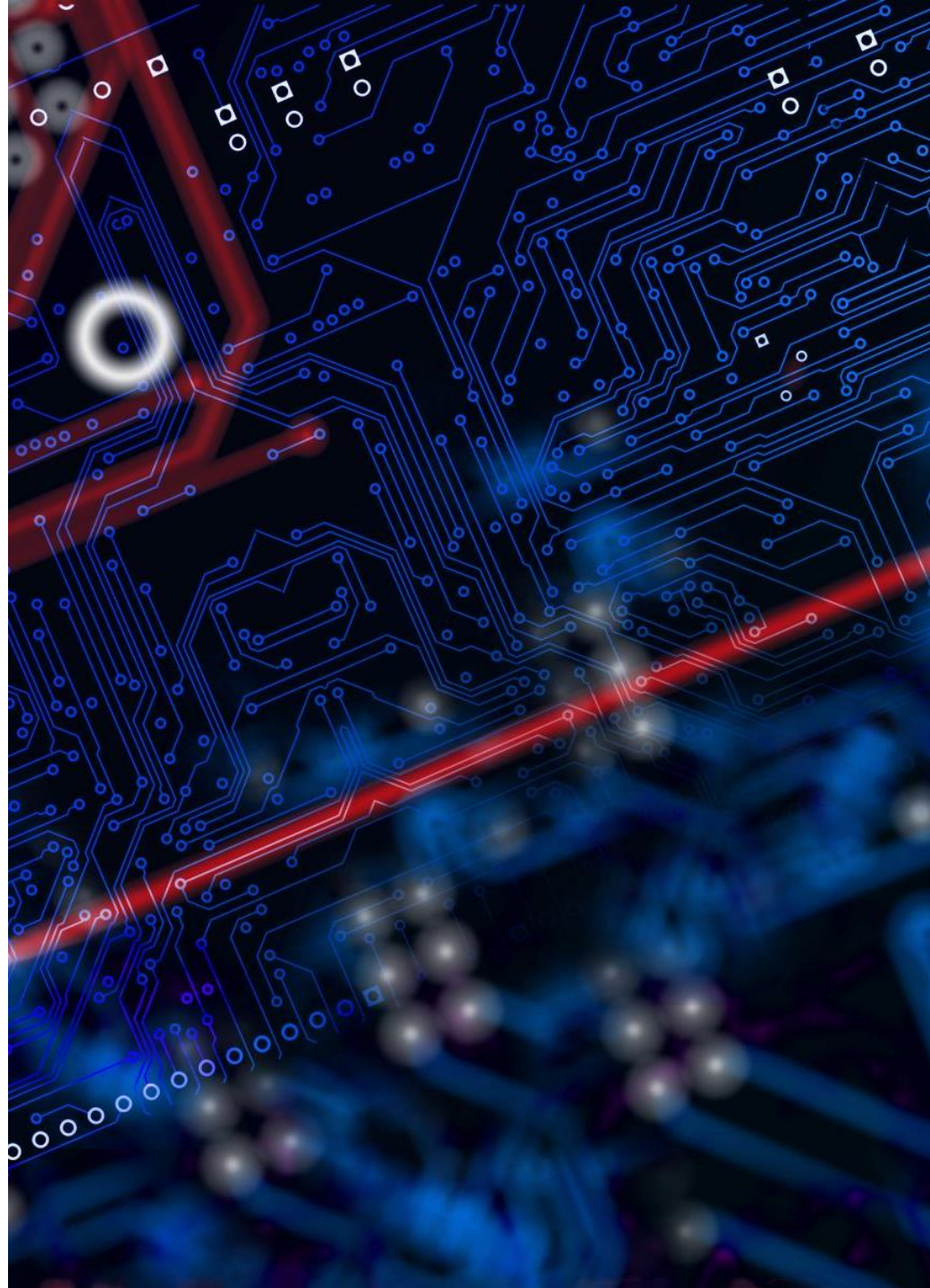

LESSON 2

BOOK CHAPTER 22

ELECTRIC FIELDS



Problem 9 (Book chapter 22)

In the adjacent figure shows two charged particles on an x axis: $-q = -3.20 \times 10^{-19} \text{ C}$ at $x = -3.00 \text{ m}$ and $+q = +3.2 \times 10^{-19} \text{ C}$ at $x = +3.00 \text{ m}$. What are the (a) magnitude and (b) direction (relative to the positive direction of the x axis) of the net electric field produced at point P at $y = 4 \text{ m}$?

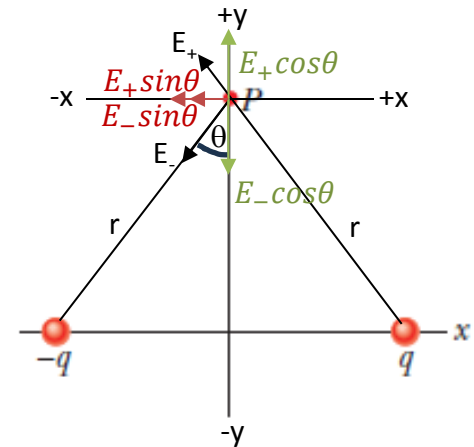
Answer:

$$E_P = E_+ \sin \theta + E_- \sin \theta$$

$$= 2E_+ \sin \theta$$

$$= 2 \times 1.15 \times 10^{-10} \times 0.6$$

$$= 1.38 \times 10^{-10} \text{ N/C}$$



$$r = \sqrt{3^2 + 4^2} = 5 \text{ m}$$

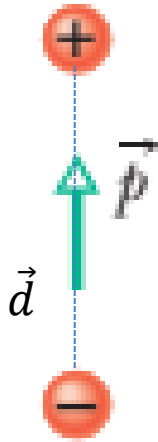
$$\sin \theta = 3/5$$

$$\begin{aligned} E_+ = E_- &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = 9 \times 10^9 \frac{3.2 \times 10^{-19}}{5^2} \\ &= 1.15 \times 10^{-10} \text{ N/C} \end{aligned}$$

Electric Dipoles

An electric dipole is a pair of point charges with equal magnitude and opposite sign (a positive charge $+q$ and a negative charge $-q$) separated by a small distance d .

Fig.1



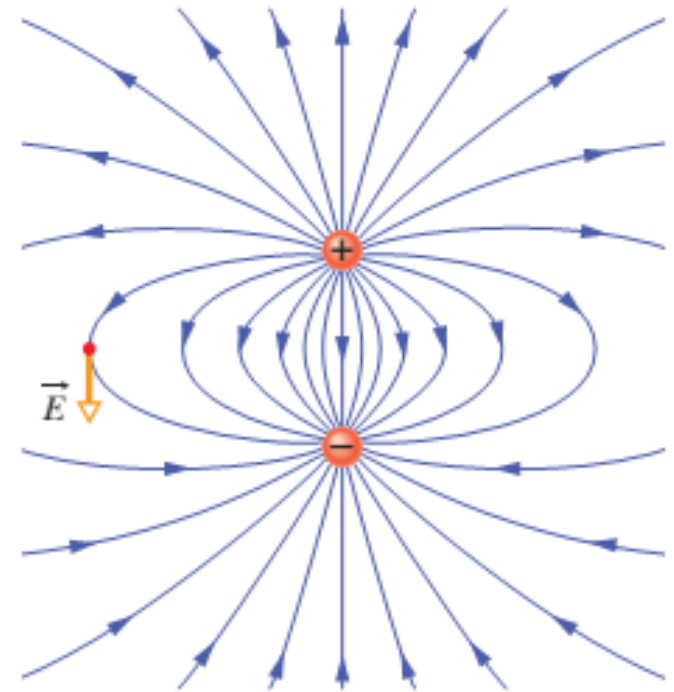
The product of the charge q and the separation d is the magnitude of a quantity called the **electric dipole moment**, denoted by p .

$$\text{That is, } p = qd$$

$$\text{In vector form, } \vec{p} = q\vec{d}$$

The direction of $\vec{p}$ is from negative charge to positive charge as shown in figure 1.

Fig.2



The pattern of electric field lines around an electric dipole, with an electric field vector $\vec{E}$ shown (Figure 2) at one point (tangent to the field line through that point).

The Electric Field Due to an Electric Dipole:

The net magnitude of the electric field at point P is

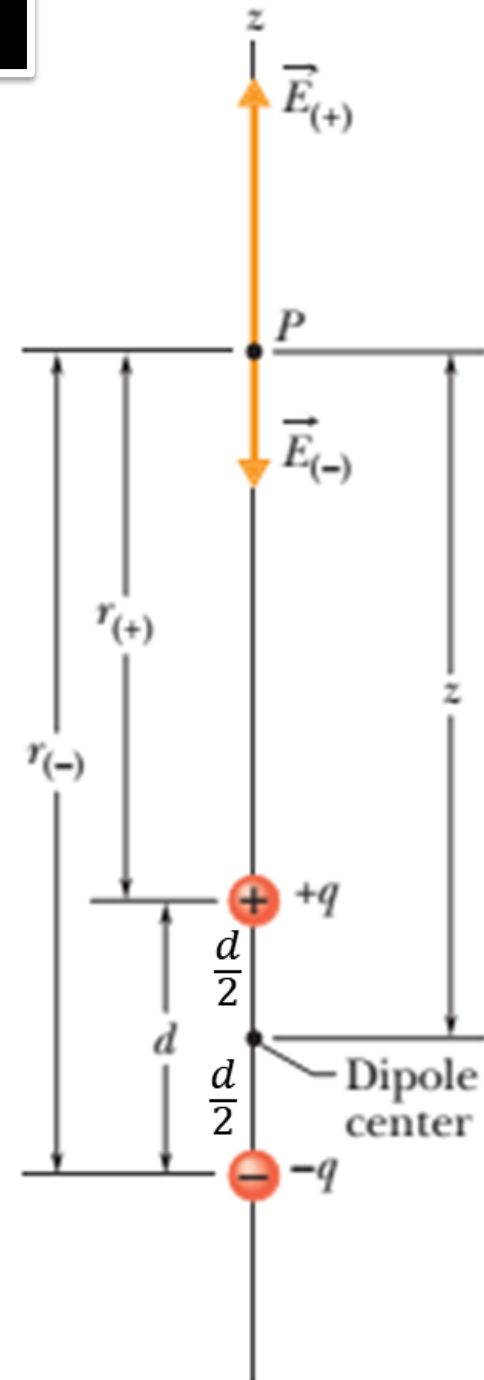
$$E = E_{(+)} - E_{(-)}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r_{(+)}^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{r_{(-)}^2}$$

$$E = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\left(z - \frac{d}{2}\right)^2} - \frac{1}{\left(z + \frac{d}{2}\right)^2} \right]$$

$$E = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\left\{z\left(1 - \frac{d}{2z}\right)\right\}^2} - \frac{1}{\left\{z\left(1 + \frac{d}{2z}\right)\right\}^2} \right]$$

$$E = \frac{q}{4\pi\epsilon_0 z^2} \left[\frac{1}{\left(1 - \frac{d}{2z}\right)^2} - \frac{1}{\left(1 + \frac{d}{2z}\right)^2} \right]$$



$$E = \frac{q}{4\pi\epsilon_0 z^2} \left[\left(1 - \frac{d}{2z}\right)^{-2} - \left(1 + \frac{d}{2z}\right)^{-2} \right]$$

For $z \gg d$, we have $\frac{d}{2z} \ll 1$

Here $\frac{d}{2z} \ll 1$

We use the form of binomial theorem,

$$(1+x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \dots \quad (x^2 < 1)$$

Therefore, we can write,

$$E = \frac{q}{4\pi\epsilon_0 z^2} \left[\left(1 + \frac{2d}{2z}\right) - \left(1 - \frac{2d}{2z}\right) \right]$$

$$(1+x)^n \approx 1 + nx \quad \text{for } x \ll 1$$

$$E = \frac{q}{4\pi\epsilon_0 z^2} \left[1 + \frac{d}{z} - 1 + \frac{d}{z} \right]$$

$$E = \frac{q}{4\pi\epsilon_0 z^2} \left[2 \frac{d}{z} \right]$$

Finally,

$$E = \frac{2qd}{4\pi\epsilon_0 z^3} = \frac{2p}{4\pi\epsilon_0 z^3} = \frac{p}{2\pi\epsilon_0 z^3}$$

here $p = qd = \text{electric dipole moment}$

Problem 7 (Book chapter 22)

In the adjacent figure, the four particles form a square of edge length $a = 5.00 \text{ cm}$ and have charges $q_1 = +10 \text{ nC}$, $q_2 = -20 \text{ nC}$, $q_3 = +20 \text{ nC}$, and $q_4 = -10 \text{ nC}$. In unit-vector notation, what net electric field do the particles produce at the square's center?

Answer:

The net electric field at the center of the square along x-axis is

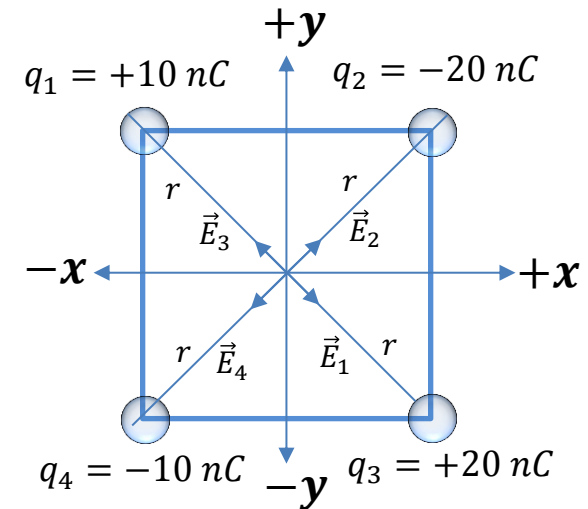
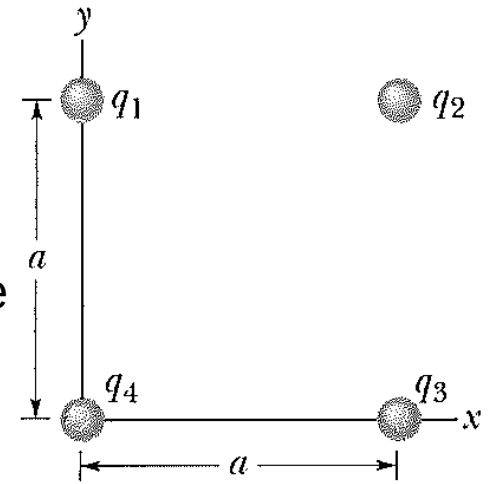
$$E_x = E_2 \cos 45^\circ + E_1 \cos 45^\circ - (E_3 \cos 45^\circ + E_4 \cos 45^\circ)$$

Here,

$$E_1 = E_4 = \frac{1}{4\pi\epsilon_0} \frac{10 \times 10^{-9}}{r^2} \text{ And } E_3 = E_2 = \frac{1}{4\pi\epsilon_0} \frac{20 \times 10^{-9}}{r^2}$$

$$E_x = E_2 \cos 45^\circ + E_1 \cos 45^\circ - E_2 \cos 45^\circ - E_1 \cos 45^\circ$$

$$E_x = 0 \text{ N/C}$$



The net electric field at the center of the square along y-axis is

$$E_y = E_2 \cos 45^\circ + E_3 \cos 45^\circ - (E_1 \cos 45^\circ + E_4 \cos 45^\circ)$$

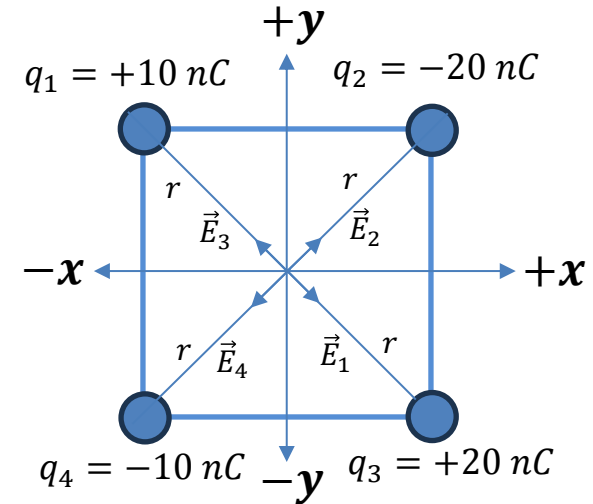
$$E_y = E_2 \cos 45^\circ + E_2 \cos 45^\circ - (E_1 \cos 45^\circ + E_1 \cos 45^\circ)$$

$$E_y = 2E_2 \cos 45^\circ - 2E_1 \cos 45^\circ$$

$$E_y = \frac{2 \times 9 \times 10^9 \times 20 \times 10^{-9} \times 0.707}{r^2} - \frac{2 \times 9 \times 10^9 \times 10 \times 10^{-9} \times 0.707}{r^2}$$

$$E_y = \frac{127.26}{r^2} = \frac{127.26}{(0.0354)^2} = 101.55 \times 10^3 \text{ N/C}$$

$$\begin{aligned} \vec{E} &= E_x \hat{i} + E_y \hat{j} = 0 + 101.55 \times 10^3 \hat{j} \\ &= 101.55 \times 10^3 \hat{j} \frac{\text{N}}{\text{C}} \end{aligned}$$



$$r = \sqrt{\frac{a^2}{4} + \frac{a^2}{4}} = \sqrt{\frac{2a^2}{4}}$$

$$r = \frac{a}{\sqrt{2}} = \frac{0.05}{1.414} = 0.0354 \text{ m}$$

Linear charge density:

When charge is distributed along a line (such as a long, thin, charged plastic rod), we use (the Greek letter lambda λ) to represent the charge per unit length known as **linear charge density**.

That is

$$\lambda = \frac{\text{Amount of charge distributed on the rod}}{\text{Length of the rod}} \quad [\text{For uniform linear charge density}]$$

The SI unit of λ is Coulomb/meter; simply, we use C/m.

Note:

Some Measures of Electric Charge

Name	Symbol	SI Unit
Charge	q	C
Linear charge density	λ	C/m
Surface charge density	σ	C/m ²
Volume charge density	ρ	C/m ³

Analytical problem:

For charge that is distributed uniformly over a ring, determine the net electric field at a given point on the axis of the ring (at a distance z from the center of the ring).

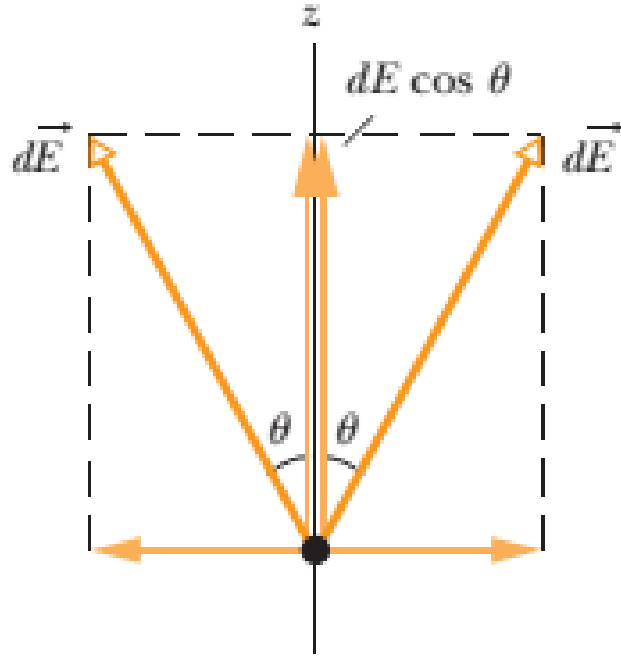
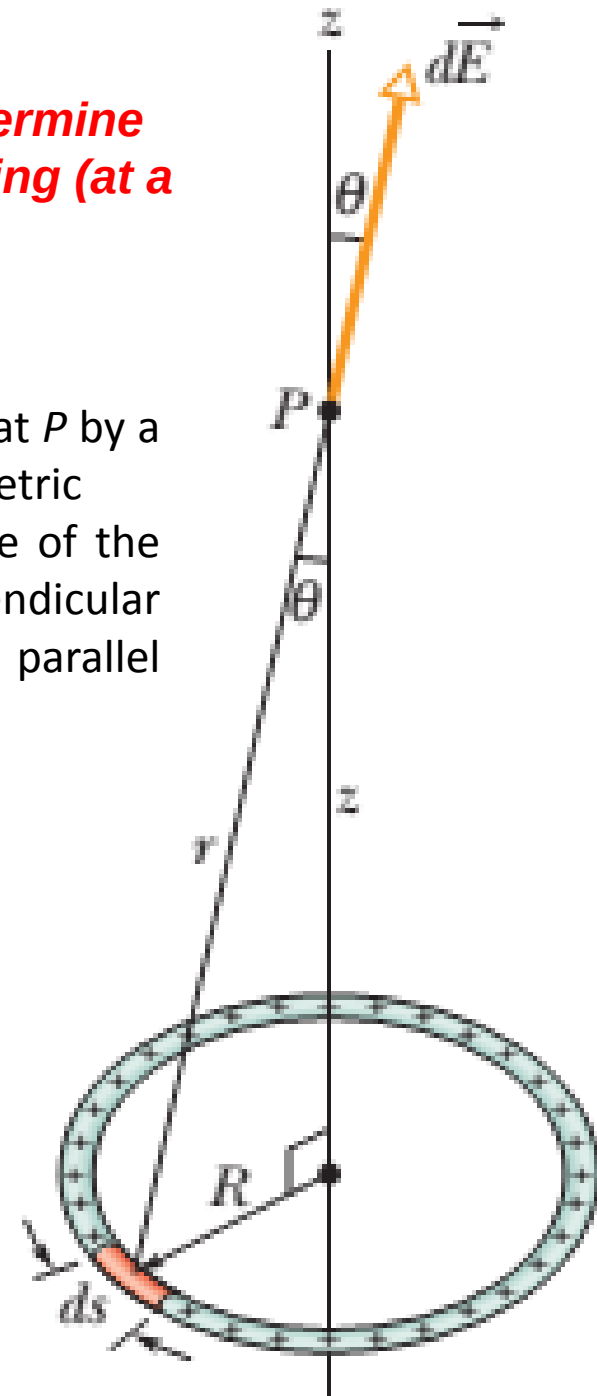


Fig. The electric fields set up at P by a charge element and its symmetric partner (on the opposite side of the ring). The components perpendicular to the z axis cancel; the parallel components add.

Let ds be the (arc) length of any differential element of the ring. Since λ is the charge per unit (arc) length, the element has a charge of magnitude

$$dq = \lambda ds$$



This differential charge (dq) sets up a differential electric field $d\vec{E}$ at point P , which is a distance r from the element. Treating the element as a point charge.

Hence, we can write the magnitude of $d\vec{E}$ as

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{(\sqrt{z^2 + R^2})^2}$$

Since the components perpendicular to the z axis cancel and the parallel components add, the net electric field along z -axis is

$$E_z = E = \int dE \cos\theta = \int \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{(z^2 + R^2)} \left[\frac{z}{\sqrt{z^2 + R^2}} \right]$$

From the figure,
we can write

$$\cos\theta = \frac{z}{r} = \frac{z}{\sqrt{z^2 + R^2}}$$

$$E = \frac{\lambda z}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} \int_{s=0}^{s=2\pi R} ds = \frac{\lambda z}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} [s]_0^{2\pi R}$$

$$E = \frac{\lambda(2\pi R) z}{4\pi\epsilon_0(z^2 + R^2)^{3/2}} = \frac{qz}{4\pi\epsilon_0(z^2 + R^2)^{3/2}}$$

Finally,

$$E = \frac{qz}{4\pi\epsilon_0(z^2 + R^2)^{3/2}}$$

Problem 30 (Book chapter 22)

Figure shows two concentric rings, of radii R and $R' = 3R$, that lie on the same plane. Point P lies on the central z axis, at distance $D = 2R$ from the center of the rings. The smaller ring has uniformly distributed charge $+Q$. In terms of Q , what is the uniformly distributed charge on the larger ring if the net electric field at P is zero?

Answer:

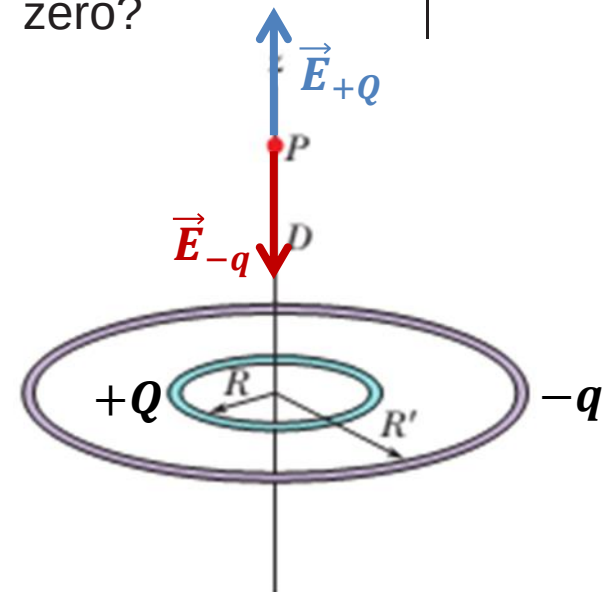
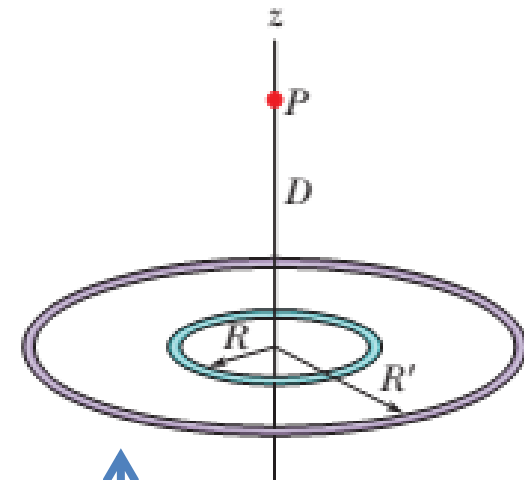
According to the statement of the problem,

$$E_{+Q} - E_{-q} = 0$$

$$E_{+Q} = E_{-q}$$

$$\frac{QD}{4\pi\epsilon_0(D^2 + R^2)^{3/2}} = \frac{qD}{4\pi\epsilon_0(D^2 + (3R)^2)^{3/2}}$$

$$\frac{Q}{(4R^2 + R^2)^{3/2}} = \frac{q}{(4R^2 + 9R^2)^{3/2}}$$



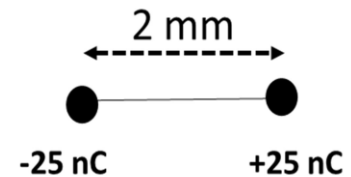
$$q = \frac{Q(13R^2)^{3/2}}{(5R^2)^{3/2}} = \left(\frac{13}{5}\right)^{3/2} Q = 4.19Q$$

That is

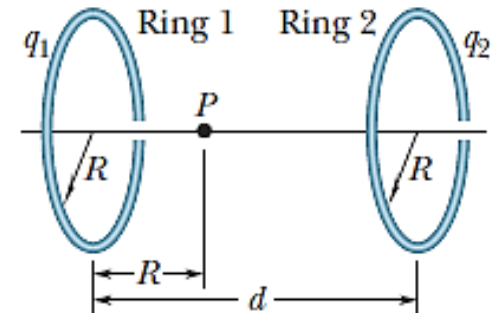
$$q = -4.19Q$$

Let's try !!!

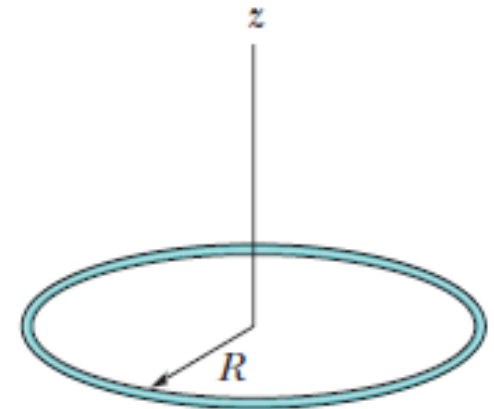
1. Calculate electric dipole moment for the adjacent figure.



2. [Problem 23]: The adjacent Figure shows two parallel nonconducting rings with their central axes along a common line. Ring 1 has uniform charge q_1 and radius R ; ring 2 has uniform charge q_2 and the same radius R . The rings are separated by distance $d = 3.00R$. The net electric field at point P on the common line, at distance R from ring 1, is zero. What is the ratio q_1/q_2 ?



3. [Problem 24]: A thin nonconducting rod with a uniform distribution of positive charge Q is bent into a complete circle of radius R . The central perpendicular axis through the ring is a z axis, with the origin at the center of the ring. What is the magnitude of the electric field due to the rod at (a) $z = 0$ and (b) $z = \infty$? (c) In terms of R , at what positive value of z is that magnitude maximum?



THANK YOU